NIMITI SHARMA

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PROFESSIONAL SUMMARY

Ambitious and research-driven B.Tech graduate specialized in Artificial Intelligence and Machine Learning with a strong academic foundation (8.05 CGPA). Proven track record in high-impact academic research, featuring a first-authored book chapter focusing on Japan's 2030 vision for 6G communication published by Taylor & Francis, a Q1-indexed journal, and an international conference paper. Experienced in cutting-edge AI domains including Deep Learning, Explainable AI (XAI), IoT integration, and Generative AI, complemented by professional peer-review experience for IEEE. Seeking to advance interdisciplinary AI research at OIST, with a focus on trustworthy and explainable AI.

RESEARCH INTERESTS

• Artificial Intelligence & Machine Learning • Deep Learning • Computer Vision • Medical Image Analysis • Explainable Artificial Intelligence (XAI) • Natural Language Processing (NLP) • Large Language Models (LLMs) & Retrieval-Augmented Generation (RAG) • Trustworthy AI for Healthcare

EDUCATION

Unitedworld Institute of Technology (Karnavati University) | Ahmedabad, India

- **Degree: B. Tech in Computer Science and Engineering (Specialization in Artificial Intelligence and Machine Learning)**
- **CGPA: 8.05 / 10.0**
- **Duration: 2021 – 2025**

RESEARCH PUBLICATIONS & BOOK CHAPTERS

- **Sharma, N.**, *et al.* (2025). **Network Security and Data Privacy in 6G Communication: Japan's 2030 Vision for 6G – A Case Study.** *Taylor & Francis*, Scopus-Indexed Book Chapter. **First Author.** 🔗
- **Sharma, N.**, *et al.* (2025). **BNS Mitra: RAG-Optimized LLM-Based AI-Powered Legal Virtual Assistant.** *Proceedings of ICCSAI 2025.* **Published.** 🔗
- **Sharma, N.**, *et al.* (2026). **Review on Air Quality Index and Air Pollutant Concentration Prediction Based on IoT and AIML Algorithms.** *Indoor Air* (Q1 Journal). **Manuscript Under Review.**

ACADEMIC SERVICE & LEADERSHIP

- **Reviewer:**
  1. IEEE Contemporary Computing Innovations Conference (CCIC), 2026.
  2. IEEE 8th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA), 2026.
  3. IEEE 11th International Conference on Research in Intelligent Computing in Engineering (RICE), 2026.
- **Secretary:** USCI Events and Media Committee, Karnavati University (2-year term).

RESEARCH EXPERIENCE & ONGOING PROJECTS

AI/ML Project Intern | Attrix Technologies (Remote) Dec 2024 – Mar 2025

- Developed and evaluated **machine learning and deep learning models** for real-world AI applications.

- Performed **data preprocessing, feature engineering, and model optimisation** to improve performance.
- Collaborated with mentors on **model deployment** and AI solution development.

Current Research Collaborations & Projects (2026):

- **Medical Imaging:** Developing GAN-based frameworks for MRI brain tumor image generation in collaboration with a researcher from Umeå University, Sweden.
- **Computer Vision:** Explainable AI and Deep Learning Framework for Multi-Class Plant Disease Detection Across 38 Crop Categories.
- **Generative AI:** Agentic AI Based Small Language Model (SLM) for Legal Support and Prompt Injection Analysis using BERT Variants.

SIGNIFICANT TECHNICAL PROJECTS

- **Smart MindAlert: Explainable AI & Deep Learning Framework for Brain Stroke CT Image Classification (Mar 2025)**
  - Developed an **AI-assisted clinical decision support framework** for automated brain stroke classification from CT images.
  - Evaluated **CNN, Xception, InceptionV3, MobileNetV2, and ResNet**, achieving **97.21% classification accuracy**.
  - Integrated **Grad-CAM, SHAP, and LIME** to provide interpretable predictions and improve model transparency.
  - Implemented a **Streamlit-based healthcare application** with emergency alerts, AI chatbot, treatment guidance, and video consultation to support early diagnosis.
- **Cutting-Edge IoT Approach to Home Safety with Gas Sensing (Mar 2024)**
  - Designed and implemented an **automated IoT-based fire detection framework** using **Raspberry Pi, OpenCV, and gas sensors** for early hazard identification.
  - Combined **computer vision and sensor-based monitoring** to enhance real-time fire prevention and emergency response capabilities.
- **Social Distancing Monitoring Using YOLOv3 (Nov 2023)**.
  - Developed a **YOLOv3-based overhead vision framework** for real-time human detection and position tracking in surveillance video streams.
  - Implemented **automated social distancing and safety boundary violation detection** through inter-person distance estimation to support intelligent public safety monitoring.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, R
- **AI, Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, CNN, DCNN, YOLO, OpenCV
- **Explainable AI (XAI) & NLP:** Grad-CAM, SHAP, LIME, NLTK, Transformers, RAG Architectures
- **Generative AI:** GANs (DCGAN, CGAN, CycleGAN), Small Language Models (SLMs)
- **Data Analysis & Visualization:** Pandas, NumPy, SciPy, Power BI, Tableau, Matplotlib, Seaborn
- **Web & Database Technologies:** Flask, Streamlit, HTML, CSS, JavaScript, MySQL, MongoDB

HONORS, AWARDS & HACKATHONS

- **Smart India Hackathon:** Secured 4th Position in University among top 10 competing teams (Mar 2022).
- **CodeUnnati Innovation Marathon:** Semi-finalist team, GTU / Edunet Foundation (Mar 2024).